Ahmed Moussaoui

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Education

Oregon State University Corvallis, OR
B.S. in Mechanical Engineering, Robotics Control Option *Expected June 2026*

Work Experience

Robotics Lab Engineer Feb. 2025 – Present
OSU College of Civil Engineering *Corvallis, OR*

- Led a 2-3 person team designing and deploying multiple automation systems for a civil engineering research lab, targeting repetitive testing processes with a 7-DOF robotic manipulator.
- Built a vision-guided pick-and-place pipeline from scratch, reducing manual technician task time by approximately 75% through automated object detection, grasp pose generation, and motion planning.
- Presented system capabilities and progress to a group of potential investors on behalf of the research group.
- Containerized the completed system using Docker for reproducible deployment across lab environments.

Undergraduate Learning Assistant, ME 203 Sept. 2025 – Present
OSU MIME Department *Corvallis, OR*

- Supported 30+ students per term in Python programming for engineering data analysis, numerical methods, and visualization.
- Led lab sessions and provided one-on-one guidance, collaborating with faculty to refine course materials.

3D Modeler May 2022 – June 2022
Freelancer, Fiverr *Remote*

- Produced manufacturing-ready CAD models for multiple clients across 3D printing, injection molding, and CNC machining. Maintained a 5-star rating across all orders.

Projects

Autonomous RC Aircraft | *ArduPilot, Mission Planner, Avionics, Fabrication* Sept. 2024 – June 2025

- Autonomous sub-team lead on a 6-10 person DBF capstone team. Built the aircraft from scratch including frame, full wiring harness, and avionics integration.
- Achieved multiple successful autonomous test flights across several test days, the first in the team's history.

SAMI Humanoid Robot – Face Display and Head Design | *ROS2, Pygame, Fusion 360, Raspberry Pi* Spring 2025

- One of 3-5 person team. Designed and fabricated the full robot head enclosure and screen mount in Fusion 360. Built an animated face display with real-time ROS2 emotion state control.
- Delivered a successful live demo with continuous reliable operation in front of an audience.

Automated Security Door System | *C++, Arduino, Fusion 360, 3D Printing* OSU ME 351

- Designed and 3D printed the full mechanical enclosure, integrating servo actuation, linkages, and sensors.
- Built a custom metal detector circuit from scratch and wrote a 5-state C++ control system with jam detection, forced entry detection, and manual override.

Technical Skills

CAD and Design: SolidWorks, Fusion 360, Autodesk Inventor, GD&T, DFM, Part and Assembly Design
Fabrication: 3D Printing (FDM), Rapid Prototyping, Machining, Soldering, Wiring and Electrical Assembly, Machine Shop
Robotics: ROS2, MoveIt2, ArduPilot, Mission Planner, Motion Planning, Systems Integration, Embedded Systems
Programming: Python, C++, MATLAB, Arduino, OpenCV, YOLO, Docker, Linux, GitHub
AI and ML: AI API Integration, Computer Vision (OpenCV, YOLO), Pre-trained Model Deployment, Model Fine-tuning
Platforms: Linux, Raspberry Pi, Raspberry Pi OS

Leadership

Head Coach, Esports Club Sept. 2024 – Apr. 2025
Oregon State University *Corvallis, OR*

- Led and managed approximately 20 players across two competitive Valorant teams, improving on-field coordination by 50% through targeted communication and teamwork drills.

Vice President, LEGO Club Nov. 2024 – Apr. 2025
Oregon State University *Corvallis, OR*

- Co-founded the club with faculty support, grew membership to 20+ students in the first term through weekly build meetings and hands-on engineering activities.